Udbhav Narayan Sharma

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ABOUT ME

M.Sc. AI & ML student at **TU Darmstadt** with **3.5 years of professional experience** in Business Analytics across FinTech and Travel industries. Currently focussed on hands-on implementation of **modern agentic AI systems**, especially the SLMs, with reduction in **hallucination**. Experience across the **ML stack** — from data analysis & preprocessing, to probabilistic model training, to on-device LLMs. Eager to apply this blend of analytics and AI research to modern ML solutions.

WORK EXPERIENCE

Assistant Manager - Planning & Growth (Data Analytics) May 2024 – Aug 2025
Cleartrip Private Limited (Flipkart) *Bangalore, India*

- **Led a team of two** and scheduled analytical systems like a **Market Dispersion Analysis** to help the Business Development teams identify targeted action items, leading to competitive pricing, enhanced hotel-end discounts, and better loyalty offers for customers.
- Analytically optimized convenience fee and coupon discounts, resulting in a **Contribution Margin (CM) growth** from below -6% to over -3%, and **automated** the process.
- Generated business analyses to drive growth in hotel bookings, resulting in a **50% increase in bookings** compared to last year same season.

Senior Business Analyst July 2022 – May 2024
Freecharge Payment Technologies Private Limited *Gurugram, India*

- Built a Machine Learning model to predict **likely loan defaulters** at 90% accuracy and **reduced the EMI default rate by 30%**.
- Drove merchant **GMV growth** through data analysis, resulting in a **60% increase** compared to the prior years' average.
- Data analysis for Merchant Cash in Advance (MCA) loan collection, leading to a **10% increase** in the percentage of due amount collected.
- Received the **FC Warrior award** for exceptional performance in the time period of November, 2022 to March, 2023 at Freecharge.

PROJECTS

Gliimr - On-device Personal Health Intelligence App | *(Solo, End-to-End)* May 2026

- Architected an **on-device**, React-native app that combines LLMs with a Dynamic Bayesian Network (DBN) resulting in a Causal Inference and Explainable AI (XAI) framework, which structurally **prevents hallucinations**.
- Engineered a **staged probabilistic-neural agentic pipeline**, with a sequential NLU pipeline with **confidence-gating**, a DBN, and multiple **ReAct** agents for sequential generation.
- Implemented a **custom in-process MCP tool dispatcher**, that handles a possible format mismatch by the generative model.
- Built a **hybrid persistence layer** using sqlite-vec for semantic memory retrieval with **recency-weighted cosine similarity**, and SQLite for structured DBN state - zero data egress.

Diabetic Retinopathy Detection July 2025

- Automated classification of Diabetic Retinopathy using a hybrid **CNN-DNN** architecture achieving **97% accuracy** and **3% False Negative Rate**, prioritizing clinical sensitivity.
- Improved model convergence by applying **CLAHE** (Contrast Limited Adaptive Histogram Equalization) preprocessing to enhance local contrast in retinal microaneurysms.
- Systematic pre-trained model selection via data augmentation and **one-epoch benchmarking**; trained final model using **OneCycleLR scheduling** on **MobileNet**.

EDUCATION

Technische Universität Darmstadt — Oct 2025 – Present
MSc. in Artificial Intelligence & Machine Learning — *Darmstadt, Germany*
Grade: 2.1 (Current) | German scale (1.0 = Best)

University of Petroleum and Energy Studies — July 2018 – July 2022
B. Tech. (Honors) in Computer Science and Engineering (Big Data) — *Dehradun, India*
Grade: 8.41/10

TECHNICAL SKILLS

AI & ML: Machine Learning, Deep Learning (DNNs, Transfer Learning), Convolutional Neural Networks (CNNs), Computer Vision
Generative AI Systems: Large Language Models, Agentic Frameworks (ReAct), RAG / Vector DB
Programming & Data: SQL, Python, MS Office (Excel), Power BI, Pandas, NumPy, PyTorch, TensorFlow, PySpark, Thematic Analysis, Time-Series Analysis

EVENTS & ACHIEVEMENTS

Open Source Contribution to Haystack | *PyCon DE 26 Sprint*
- Enhanced the DSPy-Haystack integration adapter using model-adaptive prompt templates for improved model performance.

QHack26 | *Students — General ML | Q-Summit 2026, Mannheim*
- Selected amongst 150 participants at QHack26 (General ML track).

LANGUAGES

- English – C1
- German – A2 (B1 in Progress)